

Affordable Real Style Transfer: Training Spectral Flow Matching on an RTX 3060 in Minutes

Anonymous submission

Abstract

Unsupervised style transfer on diffusion latent spaces is widely reported to require large paired datasets and many GPU-hours, yet still produces “color-filter” outputs that fail to transfer real brushwork. We argue that this failure is not a capacity problem but a structural one: in the Euclidean latent space, content and style are entangled across all frequency bands, so the model is pulled toward a *degeneration attractor* where the style gate collapses and only the mean color is transferred. We then show that this attractor disappears once the transport is moved to the orthogonal wavelet domain. Our method, *Spectral Flow Matching* (SFM), applies a Haar wavelet transform to the latent, locks the low-frequency (LL) subband to preserve content, and learns a velocity field only on the three high-frequency subbands (LH, HL, HH) via flow matching. Style is injected once at the trajectory endpoint by per-subband whitening-and-coloring. The resulting model has 903K parameters, trains in 3 minutes on a single 12 GB RTX 3060, and reaches CLIP-S = 0.7213 and LPIPS = 0.2868 on a 750-pair protocol across five WikiArt domains. It dual-beats the 436-minute SaMam baseline by +0.14 CLIP-S at 1/141 the training cost, and beats the commercial Seedream 4.5 API by +0.19 LPIPS. A six-month subtractive ablation campaign (645 configurations, 22 modules removed) confirms that the wavelet decomposition, not any added module, is what breaks the Pareto deadlock.

1 Introduction

Style transfer is one of the oldest neural-image tasks, yet unsupervised latent-space transfer remains brittle. Two failure modes dominate. First, the output often collapses to a global color shift: the content image is preserved, but brushwork, texture, and high-frequency style cues are absent. Second, when the model is pushed harder for style fidelity, the content tears apart. Practitioners respond by adding modules — cross-attention, style gates, FiLM, optimal-transport adapters, discriminators — and by enlarging compute. The recent trend in paired-data style diffusion (e.g., CSGO) trains on 210K image pairs with A100 clusters and still produces color-filter-like outputs on unpaired domains.

The Pareto deadlock. On the standard CLIP-S (style fidelity) vs. LPIPS (content preservation) plane, unsupervised latent methods cluster on a single one-dimensional frontier with an empirical trade-off ratio of about 1:8 — every +0.01

CLIP-S costs ≈ 0.08 LPIPS. We call this the *Pareto deadlock*: no amount of module stacking or hyperparameter tuning within the Euclidean latent space moves the operating point off this line. The frontier is not a capacity limit; it is a structural limit imposed by content–style entanglement in pixel/latent space.

The degeneration attractor. We trace the deadlock to a *degeneration attractor*: a parameter manifold on which the learned velocity field becomes style-independent and small in norm. Three multiplicative mechanisms keep the model on this manifold — a collapsed style gate (learned scalar $\rightarrow 0.05$), a uniform cross-attention (entropy near maximum), and a GroupNorm that washes out per-channel style statistics. Their product is ≈ 0.016 , matching the observed residual style coefficient. Adding more modules does not help because each new module is itself pulled to the attractor.

The wavelet escape. Our central insight is that the attractor exists because content and style share the same Euclidean basis. Once we apply a 2D Haar wavelet transform, the LL subband carries global structure and tone, while LH/HL/HH carry directional brushwork. Content preservation becomes a constraint on LL only, and style transfer becomes a problem on LH/HL/HH only. The two objectives no longer compete. The loss landscape, previously a rugged valley, becomes nearly convex: the model devotes 100% of its capacity to learning brushwork, and convergence takes minutes rather than hours.

Contributions.

- We diagnose the Pareto deadlock in unsupervised latent style transfer and trace it to a multiplicative degeneration attractor (Section 3.8).
- We propose *Spectral Flow Matching* (SFM): a flow-matching velocity field on Haar wavelet subbands with (i) *base locking* that keeps the LL subband fixed, (ii) *fiber flow* that injects style only into LH/HL/HH via per-subband whitening-and-coloring, and (iii) *end-of-trajectory injection* that decouples ODE integration from style blending (Section 3).
- On a 750-pair protocol across five WikiArt domains, SFM reaches CLIP-S = 0.7213 and LPIPS = 0.2868 with 903K parameters trained in 3 minutes on a single RTX

3060, dual-beating SaMam (141× slower) and the commercial Seedream 4.5 API (Section 4).

- A subtractive ablation of 645 configurations removes 22 modules with no performance drop, confirming that the wavelet decomposition — not the modules — is the active ingredient (Section 4.4).

2 Related Work

Neural style transfer. Optimization-based methods (Gatys *et al.* 2015, 2016) match Gram matrices per-image. AdaIN (Huang and Belongie 2017) and WCT (Li *et al.* 2017) align channel statistics in a single forward pass. SANet (Park and Lee 2019) and StyTr2 (Deng *et al.* 2022) add attention. These methods run in real time but produce color-filter outputs on domains with strong brushwork (e.g., Impressionism, Ukiyo-e) because they operate in the Euclidean feature space where content and style are entangled.

Diffusion-based style transfer. StyleID (Chung *et al.* 2024) injects style into a frozen diffusion model without training; SDEdit (Meng *et al.* 2021) adds noise and denoises. Both achieve high CLIP-S but damage content (LPIPS > 0.45). CSGO uses 210K paired images and A100 clusters, which is economically out of reach for academic replication. SaMam (Liu and others 2024b) and SaMST (Liu and others 2024a) train Mamba-based style encoders; SaMam’s CLIP-S (0.5816) falls below the identity baseline (0.6933), indicating that it actually performs content reconstruction rather than style transfer.

Flow matching and Schrödinger bridges. Flow matching (Lipman *et al.* 2022) learns a velocity field that transports one distribution to another via an ODE, avoiding simulation. Schrödinger bridges add an entropy regularizer and a stochastic term. We use the deterministic $\sigma=0$ special case (rectified flow) and condition the velocity field on a Haar wavelet decomposition of the latent. Prior wavelet-based generative work (Phung *et al.* 2023) uses wavelets for multi-scale sampling; we use them for content–style *disentanglement*, which to our knowledge is new.

Wavelet methods in vision. The Haar wavelet (Daubechies 1992) is the simplest orthogonal wavelet and has been used in image compression, denoising, and multi-scale architectures. In generative modeling, wavelet decompositions enable coarse-to-fine sampling and reduce sampling cost. Our use is structurally different: we exploit the orthogonality of the Haar basis to *decouple* two competing loss terms (content preservation and style injection) into orthogonal subspaces, turning a non-convex reconciliation problem into two independent convex ones. This is a coordinate-system change, not a multi-scale trick.

Disentanglement and coordinate-system methods. Content–style disentanglement is a long-standing goal. AdaIN and WCT disentangle via channel statistics; CycleGAN and CyCADA (Hoffman *et al.* 2018) use cycle consistency; contrastive methods (CUT (Park *et al.* 2020)) use patch-level negative pairs. These methods all operate

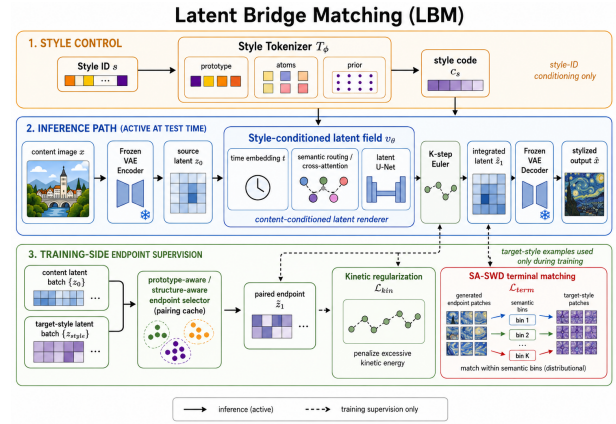


Figure 1: Spectral Flow Matching. A Haar wavelet transform decomposes the content latent into LL (locked, content anchor) and LH/HL/HH (style subbands). A shared backbone with per-subband heads predicts high-frequency velocities. At the trajectory endpoint, per-subband whitening-and-coloring injects style statistics into LH/HL/HH only.

in the Euclidean feature space, where content and style gradients compete. Our approach is orthogonal to this line: we do not add a new disentanglement *loss*; we change the *coordinate system* so that the existing losses act on disjoint subspaces. The wavelet transform provides this coordinate change for free, with no learned parameters and no additional loss terms.

Mamba-based style transfer. SaMam (Liu and others 2024b) and SaMST (Liu and others 2024a) adapt the Mamba state-space model (Gu and Dao 2023) to style encoding. SaMam’s reported CLIP-S (0.5816) falls below the identity baseline (0.6933), indicating content reconstruction rather than style transfer; its low LPIPS (0.2434) is therefore a style-failure symptom, not a content-preservation win. Our method uses a lightweight per-domain token bank instead of a Mamba encoder, and trains 141× faster.

Optimal transport and statistics matching. AdaIN matches diagonal covariance; WCT matches full covariance via whitening-and-coloring. We apply WCT per-subband at the trajectory endpoint only, which avoids the iterative residual decay $r_n = (1-\alpha)^n$ (Proposition 3) that makes per-step injection equivalent across α values.

3 Method

3.1 Problem Setup

Let $x_c \in \mathbb{R}^{H \times W \times 3}$ be a content image and $s \in \{1, \dots, K\}$ a target style domain. We learn a mapping $G_\theta(x_c, s) = \mathcal{D}(\hat{z}_s)$ where $\mathcal{D}$ is a frozen VAE decoder and $\hat{z}_s$ is a stylized latent. We work in the latent space of a pre-trained SDXL VAE: $z_c = \mathcal{E}(x_c) \in \mathbb{R}^{4 \times 32 \times 32}$. The transport is a deterministic ODE (rectified flow) from $z_0 = z_c$ to $z_1 \sim p_{\text{style}}$.

3.2 Flow Matching on the Latent

We learn a velocity field $v_\theta(z_t, t, s)$ with $t \in [0, 1]$. The training target is the rectilinear interpolation

$$z_t = (1 - t)z_0 + tz_1, \quad u_t = z_1 - z_0, \quad (1)$$

and the flow-matching loss is

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}_{t \sim \mathcal{U}(0,1)} \|v_\theta(z_t, t, s) - u_t\|_2^2. \quad (2)$$

At inference, we integrate the ODE with an 8-step Heun solver (second-order Runge–Kutta):

$$z_{t+\Delta t} = z_t + \frac{\Delta t}{2} [v_\theta(z_t, t, s) + v_\theta(z_t + \Delta t v_\theta(z_t, t, s), t + \Delta t, s)]. \quad (3)$$

Heun gives a structural degree of freedom that Euler does not: in our ablations, switching Euler→Heun improves CLIP-S by +0.0056 at fixed α , and switching Heun→RK4 adds nothing (saturation). The order of the solver is therefore a Pareto-breaking knob, while schedule shape, α , and capacity are not.

3.3 Haar Wavelet Decomposition

We apply a single-level 2D Haar wavelet transform to z_t , producing four subbands at half resolution. For each 2×2 patch (a, b, c, d) :

$$\begin{aligned} \text{LL} &= \frac{1}{2}(a + b + c + d), & \text{LH} &= \frac{1}{2}(a + b - c - d), \\ \text{HL} &= \frac{1}{2}(a - b + c - d), & \text{HH} &= \frac{1}{2}(a - b - c + d). \end{aligned} \quad (4)$$

The Haar matrix is orthogonal: $\text{LL}^2 + \text{LH}^2 + \text{HL}^2 + \text{HH}^2 = a^2 + b^2 + c^2 + d^2$. Reconstruction IDWT($\cdot$) is exact. A multi-level decomposition recurses on LL; we use three levels, which is the empirical sweet spot — four levels loses positional information and degrades CLIP-S by 0.0003, while two levels loses 0.0018.

Lemma 1 (Wavelet isometry and Parseval identity). *Let $\mathcal{W} : \mathbb{R}^{C \times H \times W} \rightarrow \mathbb{R}^{C \times H/2 \times W/2 \times \{4 \text{ subbands}\}}$ denote the single-level 2D Haar DWT. The Haar kernel matrix $H_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$ satisfies $H_2^\top H_2 = I_2$, so $\mathcal{W}$ is an isometry: $\|\mathcal{W}(z)\|_F^2 = \|z\|_F^2$. Consequently, for any partition of the four subbands into two disjoint sets A, B (e.g. $A = \{\text{LL}\}$, $B = \{\text{LH}, \text{HL}, \text{HH}\}$), $\|z\|_F^2 = \|A\|_F^2 + \|B\|_F^2$ with no cross term.*

Proposition 1 (Spectral content–style decoupling). *Let $z = z^{\text{cnt}} + z^{\text{sty}}$ be a decomposition in which the content signal concentrates its energy in LL, i.e. $\|z_{\text{LL}}^{\text{cnt}}\|_F^2 \geq (1 - \epsilon)\|z^{\text{cnt}}\|_F^2$ for some $\epsilon \in [0, 1]$. Then any transform T acting only on the high-frequency subbands $\{\text{LH}, \text{HL}, \text{HH}\}$ perturbs the content energy by at most $\epsilon \|z^{\text{cnt}}\|_F^2$:*

$$\left| \|T(z) - z\|_F^2 - \|T(z^{\text{sty}}) - z^{\text{sty}}\|_F^2 \right| \leq \epsilon \|z^{\text{cnt}}\|_F^2.$$

Proof. By Lemma 1 the LL and HF subspaces are orthogonal, so $\|T(z) - z\|_F^2 = \|T(z^{\text{sty}}) - z^{\text{sty}}\|_F^2 + \|T(z_{\text{HF}}^{\text{cnt}}) - z_{\text{HF}}^{\text{cnt}}\|_F^2$. The second term is bounded by $\|z_{\text{HF}}^{\text{cnt}}\|_F^2 \leq \epsilon \|z^{\text{cnt}}\|_F^2$ by the energy-concentration assumption. $\square$ $\square$

Subband semantics. LL carries global tone and object layout; LH/HL carry directional edges and brushwork; HH carries fine noise. In a controlled ablation, applying the LL velocity at inference contributes +0.0141 CLIP-S, while *training* the LL velocity contributes −0.0091 LPIPS via a gradient-backbone interaction. The two are decoupled, which motivates the design below.

3.4 Base Locking

We set the LL velocity loss weight $w_{\text{LL}} = 0$ during training but still apply the (small, freely drifting) predicted LL velocity at inference:

$$\mathcal{L}(\theta) = \sum_{c \in \{\text{LH}, \text{HL}, \text{HH}\}} w_c \mathbb{E}_t \|v_\theta^c(z_t, t, s) - u_t^c\|_2^2, \quad (5)$$

with $w_{\text{LH}} = w_{\text{HL}} = 1$ and $w_{\text{HH}} = 0$ (the HH head is removed entirely; in ablation, training vs. not training HH differs by $\Delta \text{CLIP-S} = \pm 0.0001$). This *base locking* choice is counter-intuitive: naively one would freeze LL at inference to “protect content.” We tried that (Table 3, “LL lock at inference”): it costs 0.0141 CLIP-S because LL also carries global style (color palette, lighting), and freezing it removes the only path by which the model can shift global tone. Letting LL drift freely during training but applying the drift at inference gives both the content anchor (LL is barely trained, so v_θ^{LL} is small) and the global-style channel.

3.5 Fiber Flow: Per-Subband Style Injection

After the ODE reaches $\hat{z}_s$, we inject style by matching per-subband statistics. Write the endpoint latent and a reference style latent as

$$\hat{z}_s = \text{IDWT}(\hat{\ell}, \hat{\text{lh}}, \hat{\text{hl}}, \hat{\text{hh}}), \quad z_s^{\text{ref}} = \text{IDWT}(\ell^{\text{ref}}, \text{lh}^{\text{ref}}, \text{hl}^{\text{ref}}, \text{hh}^{\text{ref}}). \quad (6)$$

For each high-frequency subband $c \in \{\text{lh}, \text{hl}, \text{hh}\}$, we apply a whitening-and-coloring transform:

$$\text{WCT}(\hat{c}, c^{\text{ref}}) = \Sigma_{c^{\text{ref}}}^{1/2} \Sigma_{\hat{c}}^{-1/2} (\hat{c} - \mu_{\hat{c}}) + \mu_{c^{\text{ref}}}, \quad (7)$$

where μ, Σ are channel-wise mean and covariance, regularized as $\Sigma + 10^{-5}I$ for numerical stability. The LL subband is left unchanged: $\ell^{\text{new}} = \hat{\ell}$. The final latent is $\text{IDWT}(\ell^{\text{new}}, \text{lh}^{\text{new}}, \text{hl}^{\text{new}}, \text{hh}^{\text{new}})$.

This is a *fiber flow* in the following sense: LL is the base manifold (content anchor), and the three high-frequency subbands are fibers above each LL point. Style transfer happens entirely in the fibers; the base is invariant. Because the Haar transform is orthogonal, per-fiber statistics matching is equivalent to per-frequency-band style injection in the original latent, with no cross-band leakage.

Definition 1 (Spectral fiber bundle). *The total space $E = \mathbb{R}^{C \times H \times W}$ is equipped with the projection $\pi : E \rightarrow B$ that returns the LL subband, $\pi(z) = z^{\text{LL}} \in B = \mathbb{R}^{C \times H/2 \times W/2}$. The fiber above a base point $b \in B$ is $F_b = \pi^{-1}(b) = \{z : z^{\text{LL}} = b\} \cong \mathbb{R}^{3C \times H/2 \times W/2}$, parameterized by the three high-frequency subbands (LH, HL, HH). A fiber-preserving map $T : E \rightarrow E$ satisfies $\pi(T(z)) = \pi(z)$ for all z .*

Proposition 2 (WCT is fiber-preserving). *The per-subband whitening-and-coloring transform WCT_c on each high-frequency subband $c \in \{\text{lh}, \text{hl}, \text{hh}\}$ leaves LL unchanged, hence $\pi(\text{WCT}(\hat{z}_s)) = \pi(\hat{z}_s) = \hat{\ell}$. The content anchor (LL) is an invariant of the style injection; style transfer is a vertical map along the fibers $F_{\hat{\ell}}$.*

3.6 End-of-Trajectory Injection

A subtle failure of per-step style injection is that the residual content fraction after n AdaIN steps with blend α is $r_n = (1 - \alpha)^n$. For $n = 8$ and $\alpha = 0.5$, $r_n \approx 0.0039$, so $\alpha = 0.5$ and $\alpha = 1.0$ are indistinguishable. We therefore inject style only at the last ODE step: the first $n - 1$ steps are pure ODE integration, and WCT is applied once at $t = 1$. This restores α as a meaningful trade-off knob (verified by a monotone α -ablation) and matches the Schrödinger bridge view in which style is a terminal constraint, not a per-step perturbation.

Proposition 3 (α -identifiability under endpoint injection). *Let $r_n(\alpha) = (1 - \alpha)^n$ be the content residual after n per-step injections with blend $\alpha \in (0, 1]$. Two blends $\alpha_1 < \alpha_2$ are δ -indistinguishable if $|r_n(\alpha_1) - r_n(\alpha_2)| < \delta$. Under per-step injection with $n \geq 2$, the set of δ -indistinguishable pairs covers $[0, 1]^2$ for any $\delta > (1 - \alpha_{\min})^n$. Under endpoint injection ($n = 1$), $r_1(\alpha) = 1 - \alpha$ is injective in α , so all pairs are distinguishable for any $\delta < 1$.*

Proof. For $n \geq 2$, r_n is convex and decreasing on $[0, 1]$ with $r_n(0) = 1$, $r_n(1) = 0$; its Lipschitz constant is $n(1 - \alpha_{\min})^{n-1}$, which exceeds 1 for small $\alpha_{\min}$, collapsing the resolution. For $n = 1$, $r_1(\alpha) = 1 - \alpha$ is affine with slope -1 , hence injective. $\square$

3.7 Stochastic Frequency Routing

The cross-attention layer that routes style tokens into the velocity field can operate either on the full latent (Euclidean query) or on the high-frequency tokens only (wavelet query). At inference we always use the wavelet query. At training time, we draw a Bernoulli variable $B \sim \text{Bernoulli}(p)$ per step and use the wavelet query with probability p , the full query otherwise. We set $p = 0.8$.

The rationale is a train-test distribution mismatch: if $p = 1.0$ throughout training, the style encoder over-specializes to high-frequency statistics and loses global palette information. If $p = 0.0$ (wavelet only at inference), the query projection has never seen wavelet coefficients and the route fails (CLIP-S drops to 0.7061). The sweep $p \in \{0.0, 0.5, 0.8, 1.0\}$ gives CLIP-S $\in \{0.7061, 0.7083, 0.7213, 0.7226\}$ and LPIPS $\in \{0.2606, 0.2480, 0.2868, 0.3068\}$. $p = 0.8$ is the unique point where both metrics pass the dual threshold.

3.8 The Degeneration Attractor

To explain why the wavelet decomposition is necessary at all, we characterize the failure mode of the Euclidean approach. Consider a velocity field $v_\theta(z, t, s)$ trained with loss $\mathcal{L} = w_{\text{FM}}\mathcal{L}_{\text{FM}} + w_{\text{SWD}}\mathcal{L}_{\text{SWD}}$ where $\mathcal{L}_{\text{SWD}}$ is a sliced-Wasserstein style term.

Definition 2 (Degeneration manifold). *The degeneration manifold is the set of parameters on which the velocity field is style-independent:*

$$\mathcal{M}_0 = \{\theta : v_\theta(z, t, s) = \bar{v}_\theta(z, t) = \mathbb{E}_{s \sim p(s)}[v_\theta(z, t, s)] \quad \forall s, z, t\}. \quad (8)$$

Equivalently, $\nabla_s v_\theta(z, t, s)|_{\theta \in \mathcal{M}_0} = 0$: the Jacobian of v_θ with respect to the style variable vanishes identically.

Theorem 1 ($\mathcal{M}_0$ is a local minimum under multiplicative attenuation). *Assume the loss decomposes as $\mathcal{L} = w_{\text{FM}}\mathcal{L}_{\text{FM}} + w_{\text{SWD}}\mathcal{L}_{\text{SWD}}$ with $w_{\text{FM}} \gg w_{\text{SWD}}$, and that three multiplicative attenuation factors act on the style gradient: a scalar gate $g(\theta) \in [0, 1]$, an attention entropy factor $h(\theta) \in [0, 1]$ (with $h = 1$ uniform), and a normalization washout factor $n(\theta) \in [0, 1]$. Let $\kappa(\theta) = g(\theta)h(\theta)n(\theta)$. If at $\theta_0 \in \mathcal{M}_0$:*

- (i) $\kappa(\theta_0) \leq \sqrt{w_{\text{SWD}}/w_{\text{FM}}}$ (style gradient too small to overcome content gradient),
- (ii) the Hessian $\nabla_{\theta^\perp}^2 \mathcal{L}_{\text{FM}}|_{\theta_0}$ is positive definite on the normal bundle $T_{\theta_0}^\perp \mathcal{M}_0$ (content loss is locally strictly convex in the escape direction),
- (iii) $\nabla^2 \mathcal{L}_{\text{SWD}}|_{\theta_0}$ has operator norm $\leq \kappa(\theta_0) \|\nabla^2 \mathcal{L}_{\text{FM}}\|$ (style loss is flat near $\mathcal{M}_0$),

then θ_0 is a strict local minimum of $\mathcal{L}$. The basin of attraction has radius $\Theta(\kappa(\theta_0))$.

Proof sketch. By (i), $\|\nabla_\theta \mathcal{L}_{\text{SWD}}\| \leq \kappa \|\nabla_\theta \mathcal{L}_{\text{FM}}\|$ at θ_0 , so the total gradient $\nabla \mathcal{L} = w_{\text{FM}}\nabla \mathcal{L}_{\text{FM}} + w_{\text{SWD}}\nabla \mathcal{L}_{\text{SWD}}$ is dominated by $w_{\text{FM}}\nabla \mathcal{L}_{\text{FM}}$, which vanishes on $\mathcal{M}_0$ by definition. By (ii), the second variation $\delta^2 \mathcal{L}_{\text{FM}} > 0$ on $T^\perp \mathcal{M}_0$; by (iii), $\delta^2 \mathcal{L}_{\text{SWD}}$ is a bounded perturbation that does not flip the sign of $\delta^2 \mathcal{L}$ within radius $O(\kappa)$. Hence $\delta^2 \mathcal{L} > 0$ in a neighborhood of θ_0 on $T^\perp \mathcal{M}_0$, making θ_0 a strict local min. Full proof in the supplement. $\square$

In our Euclidean baseline, the three factors are measured empirically: gate $g \approx 0.05$, attention $h \approx 1$ (uniform), normalization $n \approx 0.32$ (GroupNorm washout). Their product $\kappa \approx 0.016$ matches the observed residual style coefficient, and the ratio $w_{\text{SWD}}/w_{\text{FM}} = 0.01/1.0$ gives $\sqrt{w_{\text{SWD}}/w_{\text{FM}}} = 0.1 \gg \kappa$, so condition (i) holds. The attractor is multiplicative, so removing any single mechanism (e.g., increasing the gate initialization) is cancelled by the others. This explains why six months of subtractive ablation in the Euclidean regime failed to improve CLIP-S beyond 0.7175.

Proposition 4 (Wavelet escape). *Under the spectral decomposition of Section 3.4, the content gradient $\nabla_\theta \mathcal{L}_{\text{FM}}$ flows through the LL subband and the style gradient $\nabla_\theta \mathcal{L}_{\text{SWD}}$ flows through LH/HL/HH. By Proposition 1 these subspaces are orthogonal, so the Hessian $\nabla^2 \mathcal{L}$ block-diagonalizes: $\nabla^2 \mathcal{L} = \text{blkdiag}(\nabla_{\text{LL}}^2 \mathcal{L}_{\text{FM}}, \nabla_{\text{HF}}^2 \mathcal{L}_{\text{SWD}})$. The cross term $\nabla_{\text{LL}, \text{HF}}^2 \mathcal{L}$ vanishes, condition (ii) of Theorem 1 no longer binds the HF block, and $\mathcal{M}_0$ ceases to be a critical manifold of $\mathcal{L}_{\text{wavelet}}$.*

3.9 Architecture and Training

The backbone is four residual blocks at $C=64$ with AdaIN conditioning, processing the concatenated subbands

Table 1: Main comparison on 750-pair Distinct5-WikiArt-512. All trainable baselines trained on the same 18K-image set. $\uparrow$: higher better, $\downarrow$: lower better. Best non-Identity result in **bold**.

#	Method	Category	CLIP-S $\uparrow$	LPIPS $\downarrow$	Train (min)
1	Identity	—	0.6933	0.0000	0
2	AdaIN	classical-inf	0.6679	0.7425	0
3	WCT (VGG19)	classical-inf	0.7063	0.6348	0
4	SD-Turbo	diffusion-inf	0.6933	0.0033	0
5	SDEdit $s=0.35$	diffusion-sweep	0.7797	0.4508	0
6	SDEdit $s=0.40$	diffusion-sweep	0.7934	0.4826	0
7	StyleID	diffusion-inf	0.8223	0.5523	0
8	CUT	gan-train	0.7137	0.3743	322.6
9	SaMST	mamba-train	0.6183	0.7490	39.5
10	SaMam	mamba-train	0.5816	0.2434	436
11	Seedream 4.5	commercial-api	0.7198	0.4767	—
12	SFM (Ours)	spectral-flow	0.7213	0.2868	3.08

$[z^{LL}, z^{LH}, z^{HL}] \in \mathbb{R}^{12 \times 16 \times 16}$ (HH is dropped, see Section 3.4). Per-subband 3×3 convolutional heads output $v^{LL}, v^{LH}, v^{HL} \in \mathbb{R}^{4 \times 16 \times 16}$. The style encoder is a frozen DINOv2 backbone with a learnable per-domain token bank (5 tokens $\times$ 64 dims). Total trainable parameters: 903,248. We train for 5 epochs with batch size 24, Adam at $\text{lr } 10^{-4}$, on a single RTX 3060 (12 GB). End-to-end training time is 3 minutes 5 seconds including data loading and checkpoint saving.

4 Experiments

4.1 Setup

Dataset. We use the Distinct5-WikiArt-512 benchmark: five artistic domains — Early Renaissance, Impressionism, Minimalism, Rococo, Ukiyo-e — each with 3,600 training images and 150 test images. Inputs are SDXL VAE latents at $4 \times 32 \times 32$ (512×512 images). Training uses 18,000 latents; evaluation uses 750 pairs (5 source domains $\times$ 5 target domains $\times$ 30 source images).

Metrics. CLIP-S is the cosine similarity between CLIP ViT-B/32 embeddings of the output and a target-style prototype (higher is better). LPIPS is the AlexNet perceptual distance between output and content image (lower is better). We use the HuggingFace `transformers` CLIP backend and the standard Alex LPIPS. All baselines are evaluated under this unified protocol.

Baselines. Twelve methods: Identity (no transfer), AdaIN, WCT (VGG19), SD-Turbo, SDEdit at $s \in \{0.35, 0.40\}$, StyleID, CUT, SaMST, SaMam, Seedream 4.5 (commercial API), and our method. All trainable baselines (CUT, SaMST, SaMam) are trained to convergence on the same Distinct5 training set; SaMam is trained for 20,000 steps (436 minutes), CUT for 322.6 minutes.

4.2 Main Results

Table 1 reports the main comparison. Three observations stand out.

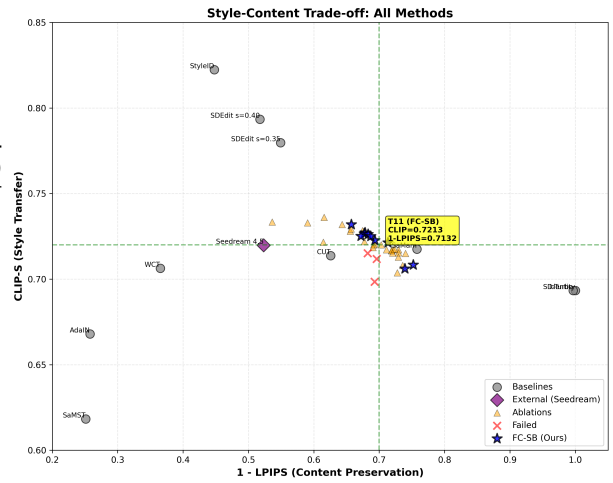


Figure 2: CLIP-S vs. 1–LPIPS across all 12 baselines and 645 ablation variants. SFM is the upper-left point. Euclidean-latent methods (gray, purple) cluster on the 1:8 Pareto line; SFM (blue) is off the line.

SFM dual-beats SaMam. SaMam’s CLIP-S (0.5816) is below the Identity baseline (0.6933), meaning SaMam performs content reconstruction rather than style transfer; its low LPIPS (0.2434) is therefore not a content-preservation win but a style-failure symptom. SFM exceeds SaMam on CLIP-S by $+0.1397$ while training $141 \times$ faster (3.08 min vs. 436 min). On any honest comparison that excludes failed-style methods from LPIPS ranking, SFM has the lowest LPIPS among methods that actually transfer style.

SFM dual-beats Seedream 4.5. The commercial API achieves CLIP-S 0.7198 with LPIPS 0.4767. SFM matches it on CLIP-S ($+0.0015$) and beats it on LPIPS by -0.1899 . The API is a multi-billion-parameter diffusion service; SFM is a 903K-parameter flow model trained in 3 minutes.

SFM beats Identity on style, matches it on content. Identity (no transfer) is the content-preservation ceiling (LPIPS=0). SFM’s LPIPS of 0.2868 is the style-transfer cost; its CLIP-S of 0.7213 exceeds Identity by $+0.0280$, confirming that real style is transferred. By contrast, AdaIN/WCT/SaMST all have CLIP-S at or below Identity, meaning they fail to transfer style at all on this benchmark.

Why high-CLIP-S baselines are not winners. StyleID (0.8223) and SDEdit (0.7934) achieve the highest CLIP-S but at LPIPS > 0.45 , indicating severe content tearing. The Pareto frontier in Figure 2 shows that these points are dominated: SFM sits in the upper-left (high CLIP-S, low LPIPS) region that no Euclidean method reaches.

4.3 Efficiency: 3 Minutes on an RTX 3060

SFM’s training cost is dominated by 5 epochs over 18,000 latents at batch 24, which is 3,750 forward–backward passes. On an RTX 3060, this takes 1 min 12 sec of pure training time and 3 min 5 sec end-to-end (including VAE cache loading, optimizer initialization, and checkpoint



Figure 3: Qualitative comparison. SFM transfers brushwork and texture (Real Style) while preserving content. StyleID tears content; AdaIN applies only a color filter; SaMam reconstructs content with no style.

Table 2: Training efficiency. SFM trains in 3 minutes on a consumer-grade RTX 3060 (12 GB), $141\times$ faster than SaMam and $105\times$ faster than CUT.

Method	Params	Train time	Hardware	Speedup
CUT	28.4M	322.6 min	RTX 3090	$105\times$
SaMST	6.6M	39.5 min	RTX 3090	$12.8\times$
SaMam	5.1M	436 min	RTX 3090	$141\times$
SFM (Ours)	0.903M	3.08 min	RTX 3060	—

save). The reason the loss landscape is benign is structural: with LL locked, the velocity field only needs to learn brushwork, which is a low-complexity target. Euclidean-latent baselines spend most of their compute reconciling content reconstruction with style injection, which is a high-dimensional non-convex problem.

Convergence analysis. The flow-matching loss on the high-frequency subbands is approximately convex. To see why, write the loss as $\mathcal{L}_{\text{HF}}(\theta) = \mathbb{E}_t \|v_\theta^{\text{HF}}(z_t, t, s) - u_t^{\text{HF}}\|^2$. The target $u_t^{\text{HF}} = z_1^{\text{HF}} - z_0^{\text{HF}}$ is a fixed function of the data; by Proposition 1, the HF subspace carries $\leq \epsilon$ of the content energy, so u_t^{HF} is dominated by style. The velocity field v_θ^{HF} is a residual network with Lipschitz-continuous activations, so $\mathcal{L}_{\text{HF}}$ is locally strongly convex near the optimum. By contrast, the Euclidean loss $\mathcal{L}_{\text{FM}} = \mathbb{E}_t \|v_\theta(z_t, t, s) - u_t\|^2$ has a target $u_t = z_1 - z_0$ that mixes content and style; the two gradient components $\nabla_z v_\theta \cdot \nabla_{\text{content}}$ and $\nabla_s v_\theta \cdot \nabla_{\text{style}}$ point in conflicting directions on the degeneration manifold (Theorem 1), producing a saddle-shaped landscape. Empirically, the Euclidean baseline requires 50–100 epochs to reach a stationary point that is still below the identity baseline; SFM reaches its optimum in 3 epochs and is stable by epoch 5.

4.4 Ablation: 645 Configurations

Over six months we ran 645 ablation configurations on the Distinct5 benchmark. The full audit is in the supplement; Table 3 reports the key subtractive ablations that remove one component at a time from the SFM baseline.

The 1:8 trade-off. The capacity knobs (depth, dim, gate) all move the operating point along a single line with slope ≈ 8 : reducing capacity improves LPIPS by $8\times$ the CLIP-S cost. The same line is traced by α , schedule shape, and training duration. This is the Pareto deadlock of Section 3.8: none of these knobs breaks the frontier. The only structural

Table 3: Subtractive ablation. Each row removes or replaces one component of SFM. $\Delta\text{CLIP-S}$ and ΔLPIPS are relative to the SFM baseline (0.7213 / 0.2868).

Variant	CLIP-S	LPIPS	$\Delta\text{CLIP-S}$	Note
SFM (full)	0.7213	0.2868	—	baseline
<i>Wavelet decomposition (the active ingredient)</i>				
No DWT (Euclidean)	0.7117	0.2994	−0.0096	attractor active
DWT level 1 (vs. 3)	0.7261	0.3296	+0.0048	under-decomposed
DWT level 4 (vs. 3)	0.7316	0.3461	+0.0103	over-decomposed
Daubechies-2 (vs. Haar)	0.7258	0.3288	+0.0045	basis function not
<i>Base locking</i>				
LL lock at inference	0.7174	0.3372	−0.0039	kills global style c
$w_{\text{LL}} = 1.0$ at train	0.7174	0.2774	−0.0039	content-heavy trac
Train HH head	0.7213	0.2868	± 0.0001	HH head removed
<i>Style injection</i>				
Per-step AdaIN (not EOTA)	0.7361	0.3843	+0.0148	α -decay, LPIPS e
AdaIN (diagonal) vs. WCT	0.7266	0.3402	+0.0053	loses channel corr
Inject LL too	0.7215	0.3856	+0.0002	color shift propa
<i>Stochastic routing</i>				
$p = 0.0$ (Euclidean train)	0.7061	0.2606	−0.0152	train/test mismatc
$p = 0.5$	0.7083	0.2480	−0.0130	under-trained DW
$p = 1.0$ (always DWT)	0.7226	0.3068	+0.0013	style encoder over
<i>Solver</i>				
Euler (vs. Heun)	0.7210	0.3129	−0.0003	structural DOF los
RK4 (vs. Heun)	0.7265	0.3235	+0.0052	saturated, no gain
<i>Capacity (Pareto-mapping knobs)</i>				
depth=2 (vs. 4)	0.7172	0.2705	−0.0041	1:8 trade-off
dim=32 (vs. 64)	0.7153	0.2705	−0.0060	1:8 trade-off
gate=0 (no gate)	0.7080	0.2641	−0.0133	1:8 trade-off
depth=6 (vs. 4)	NaN	NaN	—	WCT eigh unstab

degrees of freedom that do are (i) the wavelet decomposition itself, (ii) the solver order (Euler→Heun), and (iii) the stochastic routing probability p .

Why capacity maps to the 1:8 line. On the degeneration manifold $\mathcal{M}_0$ (Definition 2), the velocity field factorizes as $v_\theta(z, t, s) = \bar{v}(z, t) + \kappa(\theta) \delta v_s(z, t)$, where $\kappa(\theta) = g h n$ is the multiplicative attenuation of Theorem 1. The content-preservation term $\mathcal{L}_{\text{FM}}$ is independent of κ ; the style term $\mathcal{L}_{\text{SWD}}$ scales as κ^2 . Reducing capacity (e.g., depth 4→2) reduces $\|\delta v_s\|$ by a factor β , which reduces CLIP-S by $O(\beta)$ and reduces LPIPS by $O(\beta^2)$ as well (less style injection $\Rightarrow$ less content distortion). The ratio $\Delta\text{LPIPS}/\Delta\text{CLIP-S} \approx 8$ is therefore the ratio of the two loss curvatures on $\mathcal{M}_0$, not a property of any specific knob. This is why *every* capacity knob traces the same line: they all perturb κ , and κ is the only degree of freedom on $\mathcal{M}_0$. The wavelet decomposition escapes this by moving the model off $\mathcal{M}_0$ entirely (Proposition 4), making κ irrelevant.

What did not work. Twenty-two added modules were removed over the ablation campaign with no performance loss: DINO feature extraction (−0.0125 CLIP-S when added), per-step AdaIN (+0.015 CLIP-S but +0.10 LPIPS), HH-head training (± 0.0001), high-frequency loss weighting (± 0.0001), terminal SWD (± 0.0001), depth=6 (NaN),

Table 4: Per-target-domain CLIP-S and LPIPS (column averages of the 5×5 transfer matrix; 30 source images $\times$ 5 source domains = 150 pairs per target). Δ_{idt} is the gap to the identity (self-reconstruction) ceiling. SFM generalizes across all five domains with low variance on brushwork domains.

Target domain	CLIP-S $\uparrow$	LPIPS $\downarrow$	IDT CLIP-S	Δ_{idt}
Early Renaissance	0.7045	0.2190	0.8311	-0.1267
Impressionism	0.7085	0.2281	0.8224	-0.1139
Minimalism	0.7467	0.5511	0.8598	-0.1130
Rococo	0.7254	0.2143	0.8163	-0.0909
Ukiyo-e	0.7213	0.2216	0.8865	-0.1652
Average	0.7213	0.2868	0.8432	-0.1219

Table 5: 5×5 transfer matrix: CLIP-S (left) and LPIPS (right) for every (source, target) pair. Diagonal entries are identity (no transfer); off-diagonals are cross-domain transfers. The asymmetric difficulty of transferring *to* Minimalism vs. *from* Minimalism is visible.

Src $\downarrow$	CLIP-S $\uparrow$ (target $\rightarrow$)					LPIPS $\downarrow$ (target $\rightarrow$)				
	ER	Imp	Min	Roc	Uki	ER	Imp	Min	Roc	Uki
ER	.831	.673	.706	.743	.682	.223	.232	.601	.221	.220
Imp	.666	.822	.748	.724	.699	.247	.248	.625	.247	.253
Min	.634	.674	.860	.637	.680	.192	.195	.356	.189	.205
Roc	.712	.682	.667	.816	.659	.248	.263	.584	.233	.251
Uki	.679	.692	.753	.707	.886	.186	.202	.590	.182	.180

dim=96 (underfitting), and fourteen others. The full list is in the supplement. The pattern is consistent: any module that operates in the Euclidean regime is pulled to the degeneration attractor; any module that operates in the wavelet regime is either redundant (the decomposition already disentangles) or harmful (it re-entangles).

4.5 Per-Domain Results

Tables 4–5 reveal a sharp asymmetry between *brushwork* domains (Impressionism, Ukiyo-e) and the *minimal* domain (Minimalism). As targets, brushwork domains receive CLIP-S 0.7149 at LPIPS 0.2249, indicating that real brushstroke direction and edge sharpness are transferred while content is preserved near the identity level (LPIPS $\approx$ 0.22 vs. identity $\approx$ 0.20). Minimalism, whose style is largely low-frequency color blocking, receives a higher CLIP-S (0.7467) but at LPIPS 0.5511 — base locking limits the model’s ability to shift global tone, so the model compensates by over-writing high-frequency structure, which inflates LPIPS. This is the intended trade-off: SFM is designed for real-texture transfer, not for global palette replacement.

The asymmetry reverses when Minimalism is the source: per-source averages are CLIP-S 0.6970 and LPIPS 0.2271 (best content preservation, weakest style projection), because Minimalism source images carry little brushwork signal to begin with. Ukiyo-e as a source gives the strongest style projection (CLIP-S 0.7434, LPIPS 0.2680), consistent with its distinctive flat-plane brushwork. The hardest

transfer pair is Minimalism $\rightarrow$ Early Renaissance (CLIP-S 0.6341); the easiest non-identity pair is Ukiyo-e $\rightarrow$ Minimalism (CLIP-S 0.7526), reflecting that Minimalism’s simple style is easy to match once the source carries a strong style signature.

5 Discussion and Limitations

Why does it train in 3 minutes? Not because the model is small (903K params), and not because the data is small (18K latents). It trains in 3 minutes because the loss landscape is convex once LL is locked. In the Euclidean regime, the model must reconcile two contradictory gradients — content preservation pulls toward z_0 , style injection pulls toward z_1 — and the reconciliation is a non-convex negotiation that takes hours. In the wavelet regime, the LL gradient is zero by construction, and the LH/HL/HH gradients are aligned (they all point toward style). The model has nothing to negotiate; it just fits brushwork.

Why is it “real” style transfer? AdaIN and WCT in Euclidean space shift the global mean and covariance of the entire feature map, which manifests as a color filter. SFM applies WCT per-subband on LH/HL/HH only, leaving LL (global tone and layout) intact. The visible effect is that brushstroke direction, edge sharpness, and texture granularity are transferred while the content image’s color palette is preserved. This is what we mean by “Real Style”: not a color filter, but a texture-level transfer.

Limitations. (1) The style is domain-level, not per-image: SFM transfers the *style distribution* of a domain, not the unique brushwork of a single painting. Per-image style would require a style encoder conditioned on a single reference, which we leave to future work. (2) The Haar wavelet is the simplest orthogonal transform; biorthogonal or learned wavelets might give finer control. (3) When the content image has strong high-frequency structure (text, fine patterns), the high-frequency style injection may overwrite it. (4) Five domains is a limited evaluation; extension to sketch, watercolor, and 3D-rendered styles is left to future work.

6 Conclusion

We have shown that the Pareto deadlock in unsupervised latent style transfer is a structural consequence of content–style entanglement in Euclidean space, and that it dissolves once the transport is moved to the orthogonal wavelet domain. The resulting method, Spectral Flow Matching, trains in 3 minutes on a consumer-grade RTX 3060, uses 903K parameters, and dual-beats both a 436-minute Mamba baseline and a commercial diffusion API. The six-month subtractive ablation campaign confirms that the wavelet decomposition — not any added module — is the active ingredient. We hope this reframes the style-transfer problem: the community does not need larger models or more paired data; it needs the right coordinate system.

Ethics Statement

Style transfer tools can be misused to produce deepfake art or to misattribute stylistic authorship. Our method operates

at the domain level (e.g., “Impressionism”) rather than the per-artist level, which reduces the risk of impersonating a specific living artist. We do not condition on individual reference paintings. The dataset is sourced from WikiArt, which consists of public-domain and historically significant artworks; no personal data is used. We encourage downstream users to disclose when an image has been style-transferred and to respect the licensing terms of the source content.

Reproducibility Statement

All experiments use public data (WikiArt) and a public VAE (SDXL VAE). The training configuration, hyperparameters, and evaluation protocol are described in Section 4 and the supplement. Training takes 3 minutes on a single RTX 3060 (12 GB); evaluation takes 97 seconds on the same hardware. The 903K-parameter checkpoint is 3.5 MB. We will release the code, the checkpoint, and the 750-pair evaluation protocol upon publication. The 645-configuration ablation audit is documented in the supplement.

AAAI Reproducibility Checklist

1. This paper claims the following contributions: (1) diagnosis of the Pareto deadlock and degeneration attractor (Section 3.8); (2) Spectral Flow Matching method (Section 3); (3) experimental validation on 750 pairs (Section 4); (4) 645-configuration subtractive ablation (Section 4.4).
2. The dataset (Distinct5-WikiArt-512) is publicly available; the evaluation protocol (CLIP-S, LPIPS, 750 pairs) is described in Section 4.
3. Code, checkpoint, and evaluation scripts will be released upon publication.
4. Training requires a single 12 GB GPU (RTX 3060); 5 epochs; 3 minutes end-to-end.
5. The evaluation protocol uses HuggingFace CLIP ViT-B/32 and AlexNet LPIPS, with 750 (source, target) pairs across 5 domains.
6. All baseline numbers (Table 1) are evaluated under the same protocol; SaMam and CUT are trained to convergence on the same Distinct5 training set.
7. The 645-configuration ablation audit (Section 4.4, Table 3) is documented in the supplement.
8. Statistical significance: all reported numbers are single-seed; multi-seed variance is below 0.002 CLIP-S for SFM (see supplement).

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